

17-12-19

Roll No.

TCS-706

**B. TECH. (CSE)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019
ARTIFICIAL INTELLIGENCE**

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any two sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are ten.

1. Attempt any two parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Explain intelligence and artificial intelligence. How do you distinguish between the two ?

(b) Discuss the relative merits of Depth-first and Breadth-first searching methods.

(c) Write a short note on Turing Test.

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Describe in detail the difference between natural language understanding and language generation.

(b) Parse the sentence "Kathy jumped the horse" using top-down parsing.

(c) What is parsing ?

3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Explain TOP-DOWN and BOTTOM-UP parsing in detail with the help of suitable example.

(b) What do you mean by PEAS ? Determine the PEAS values for following AI problem : Vacuum Cleaner.

(c) Explain the concept of Natural Language Processing System. Explain transition Net, RTN and ATN in NLP.

4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Explain A* searching technique in detail with an example.

(3)

- (b) Derive a parse tree for the sentence "Gaurav likes the orange". where the following rules are used :

$S \rightarrow NPVP$

$NP \rightarrow N$

$NP \rightarrow DET N$

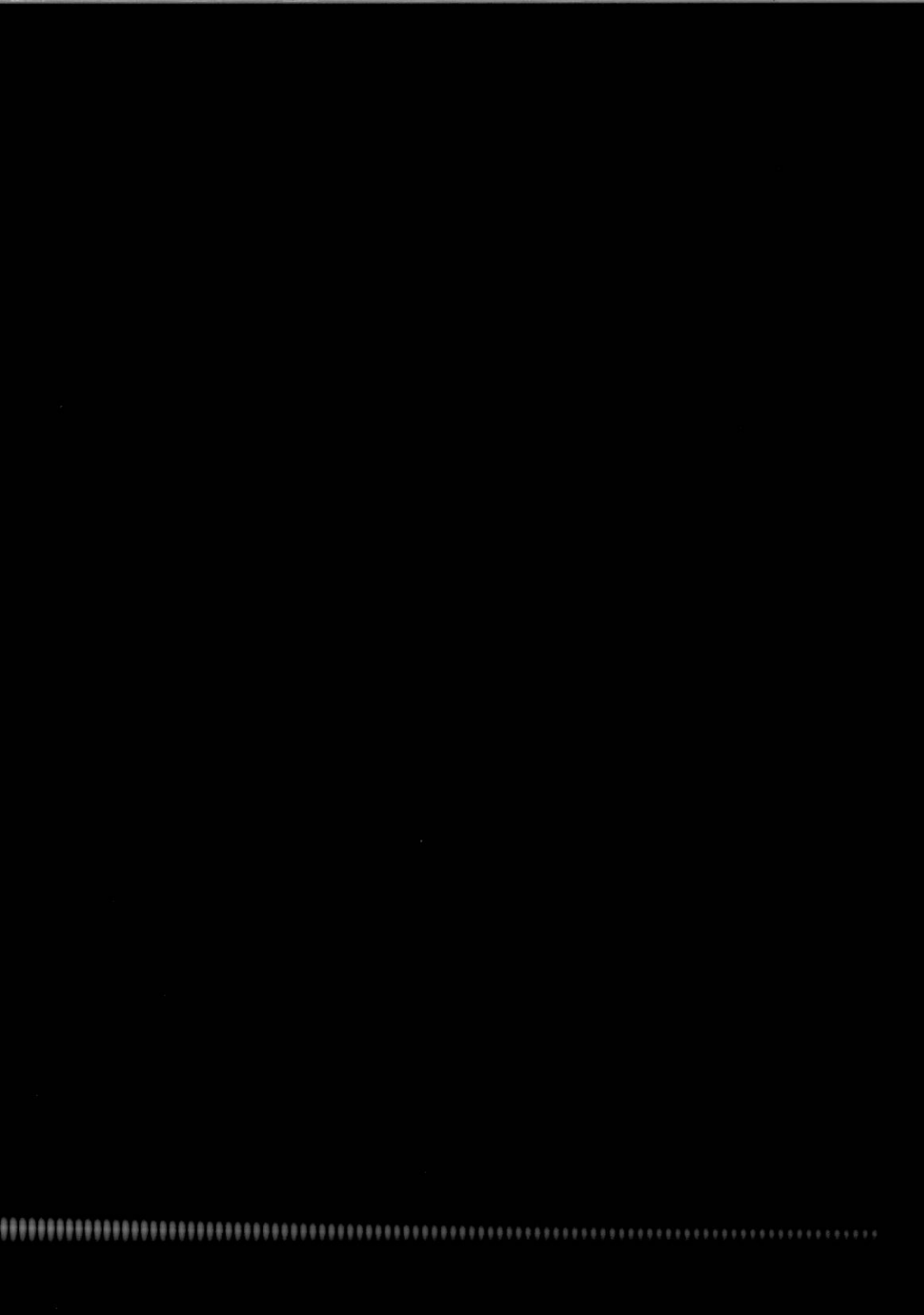
$VP \rightarrow V NP$

$DET \rightarrow \text{the}$

$V \rightarrow \text{likes}$

$N \rightarrow \text{Gaurav/Orphanage}$

- (c) What is min-max algorithm for game playing ?
5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain the concept and all phases of Natural Language Processing System.
- (b) Enumerate the Classical 'Water Jug Problem'. Describe the state space for this problem. Solve this problem by giving its opera loss sequence.
- (c) Define Rational Agent.



Roll No.

TIT-701

**B. Tech. (IT) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019**

MOBILE COMPUTING

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

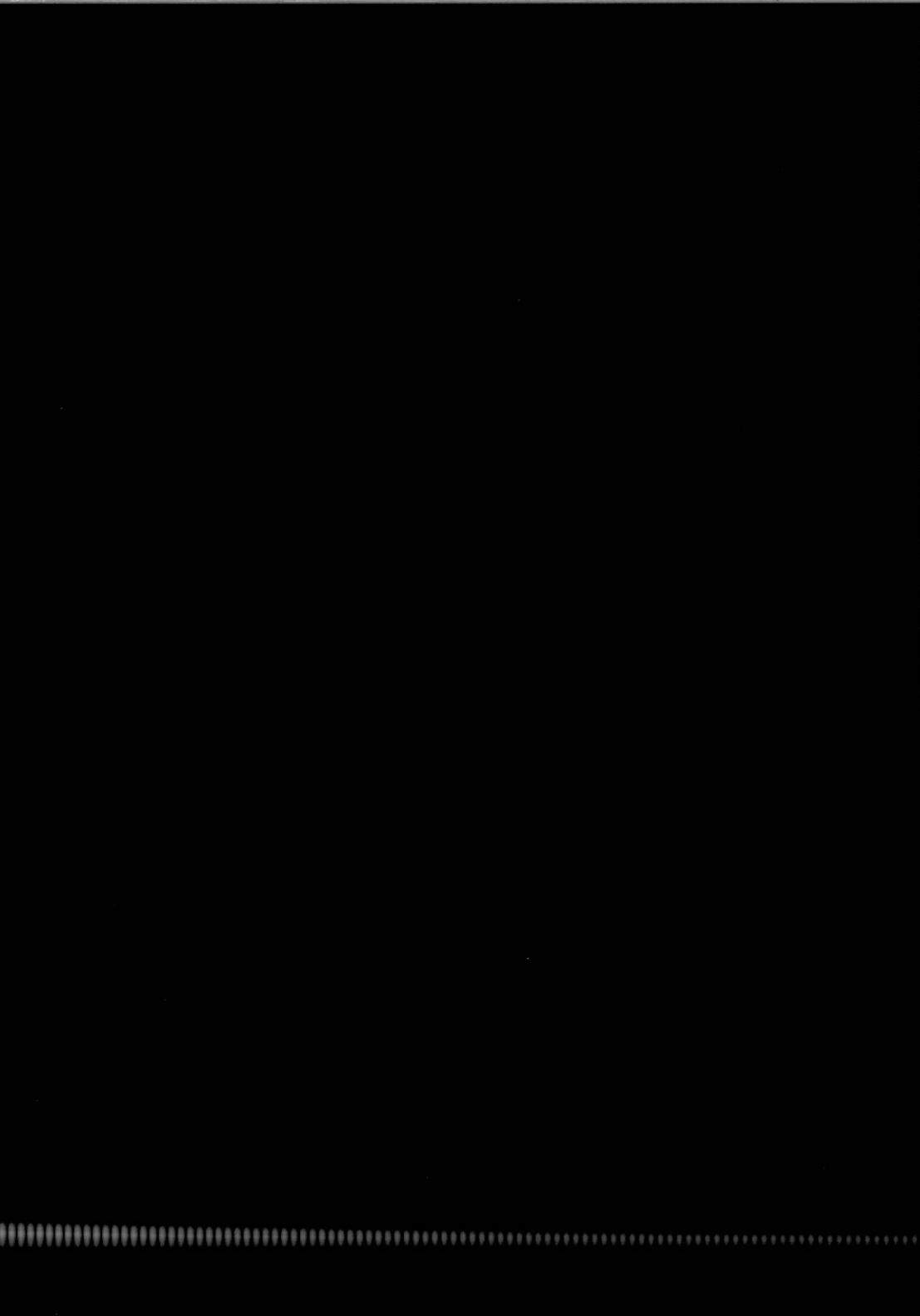
(a) Define the term Mobile Computing. List and briefly discuss the issues involved in Mobile Computing.

(b) Draw and explain the schematic diagram of cellular architecture. Do explain the entities involved.

- (c) Define : Cell, Cluster, Co-channel cells, Co-channel interference and Safe Reuse Distance in context to cellular systems.
2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) What is the difference between cell splitting and cell sectoring ? How are these related to co channel interference ?
- (b) What are Traffic Channels ? How are traffic channels allocated ?
- (c) What role do HLR and VLR play during the mobility of devices ?
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) What is GPRS ? Draw and briefly discuss its architecture.
- (b) What is Bluetooth ? Draw the Bluetooth protocol stack and describe the specific functionality of Baseband layer, RFCOMM layer and SDP layer.
- (c) Define 802.11 protocol stack with the help of a suitable diagram.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) In what Functionalities Switches differ from Routers ?

(3)

- (b) What is Mobile IP ? Explain the following terms in context to Mobile IP :
- (i) Home Agent
 - (ii) Tunneling
 - (iii) Mobility Binding
- (c) What do you understand by WAF ? Draw and explain its architectures.
5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Is standard TCF suitable for wireless networks ? Explain your viewpoint.
 - (b) What is the significance of location management in Mobile Computing ?
 - (c) What are Data Management issues in mobile computing ?



Roll No.

TCS-701

B. Tech. (IT) (SEVENTH SEMESTER) MID SEMESTER EXAMINATION, 2019

OBJECT ORIENTED ANALYSIS AND DESIGN

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) How does the object-oriented approach differ from traditional top-down approach ?

(b) What is system development methodology ?

(c) Describe the components of the unified approach.

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) What is the difference between a method and message ?
 - (b) Identify objects in a payroll system.
 - (c) Why is polymorphism useful ?
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Can you list the phases of unified process ?
 - (b) Why is reusability important ? How does object-oriented software development promote reusability ?
 - (c) How is software verification different from software validation ?
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) What is the strength of Jacobson et al. methodology ?
 - (b) Consult the web or library to obtain an article that compares different methodologies.
 - (c) Describe the difference between patterns and framework.

(3)

5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) What is the difference between sequence diagrams and collaboration diagrams ?
- (b) How would you show complete and incomplete generalizations ?
- (c) Name and describe the relationship in a use case diagram.

Roll No.

TCS-771

**B. TECH. (CSE) (SEVENTH
SEMESTER)**

MID SEMESTER EXAMINATION, 2019

**NATURAL LANGUAGE PROCESSING
USING BIGDATA**

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are ten.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) What is the difference between natural language and computer language ?

(b) Write regular expression to match the following classes of strings :

(i) A single determiner (assume that *a*, *an* and *the* are the only determiners).

- (ii) An arithmetic expression using integers, addition and multiplication, such as $2*3 + 8$.
- (c) What makes Natural Language Processing difficult ?
2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Write a note on Ambiguity and Uncertainty in Language.
- (b) Compute the edit distance (using insertion cost 1, deletion cost 1, substitution cost 1) of "sada" to "asal". Show your work (using the edit distance grid).
- (c) What are the applications of Natural Language Processing ?
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain the concept of parser. What is the role of CFG (Context-Free Grammar) in Parser ?
- (b) Write out the equation for trigram probability estimation. Now write out all the non-zero trigram probabilities for the following Corpus :
- <s> I am Sam </s>
- <s> Sam I am </s>
- <s> I do not like green eggs and ham </s>

(c) What are the different levels of Language Analysis ?

4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) What are the Elements of Semantic Analysis ?

(b) You are given the grammar below. How many parse trees can you derive for the sentences.:

Radha drove to Agra and Delhi in November :

Draw each parse tree .

The rules of the CFG grammar where S is the start symbol are :

$S \rightarrow NP VP$

$VP \rightarrow V NP | V PP | V P PP$

$NP \rightarrow NP PP | NP CNJ NP$

$PP \rightarrow P NP$

$NP \rightarrow \text{Radha} | \text{Agra} | \text{Delhi} | \text{November}$

$V \rightarrow \text{drove}$

$P \rightarrow \text{to} | \text{in}$

$CNJ \rightarrow \text{and}$

(c) How to evaluate Natural Language understanding system ?



5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Discuss various Language Models.

(b) Use the following grammar :

$S \rightarrow NP VP$ $S \rightarrow VP$ $NP \rightarrow Det Noun$

$NP \rightarrow noun$ $NP \rightarrow NP NP$ $VP \rightarrow VP NP$

$VP \rightarrow Verb$ $VP \rightarrow VPPP$ $PP \rightarrow$
preposition NP

Introduction lexicon rules of words appearing in the sentence. Give *two* possible parse tree for the sentence.

Pluck the flower with stick.

- (c) What are the *three* main techniques used for machine translation ? Very briefly say what is/are the main resource/s required in each.

Roll No.

TCS-602

**B. TECH. (CSE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, 2019**

SOFTWARE ENGINEERING

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections.

(ii) Both Sections are compulsory.

Section—A

1. Fill in the blanks/True-False : (1×5=5 Marks)

(a) The process of developing a software product using software engineering principles and methods is referred to as,

_____.
(Software evolution/Software myths)

(b) Which model is also called as the classic life cycle or the Waterfall model ?

(Iterative Development/Linear
Sequential Development)

(2)

TCS-602

(c) In the V-V model of software development, testing starts in parallel with the development. (True/False)

(d) Initially a context diagram is drawn, which is a simple representation of the entire system under investigation. This is followed by which level of diagram?

(Level1/Level2)

(e) What is the final outcome of the requirements analysis and specifications phase? (Drawing the data flow diagram/

The SRS Document)

2. Attempt any *five* parts : (3×5=15 Marks)

(Define/Short Numerical/Short Programming/ Draw)

(a) What are functional and non-functional requirements?

(b) Differentiate between Software and Program.

(c) Why is Spiral Model also called Meta Model?

(3)

TCS-602

(d) What are the different types of design? List their activities.

(e) What is maintenance? Describe its types.

(f) Discuss levels of testing.

Section—B

3. Attempt any *two* parts of choice from (a), (b) and (c). (5×2=10 Marks)

(a) Describe the software development life cycle for software development.

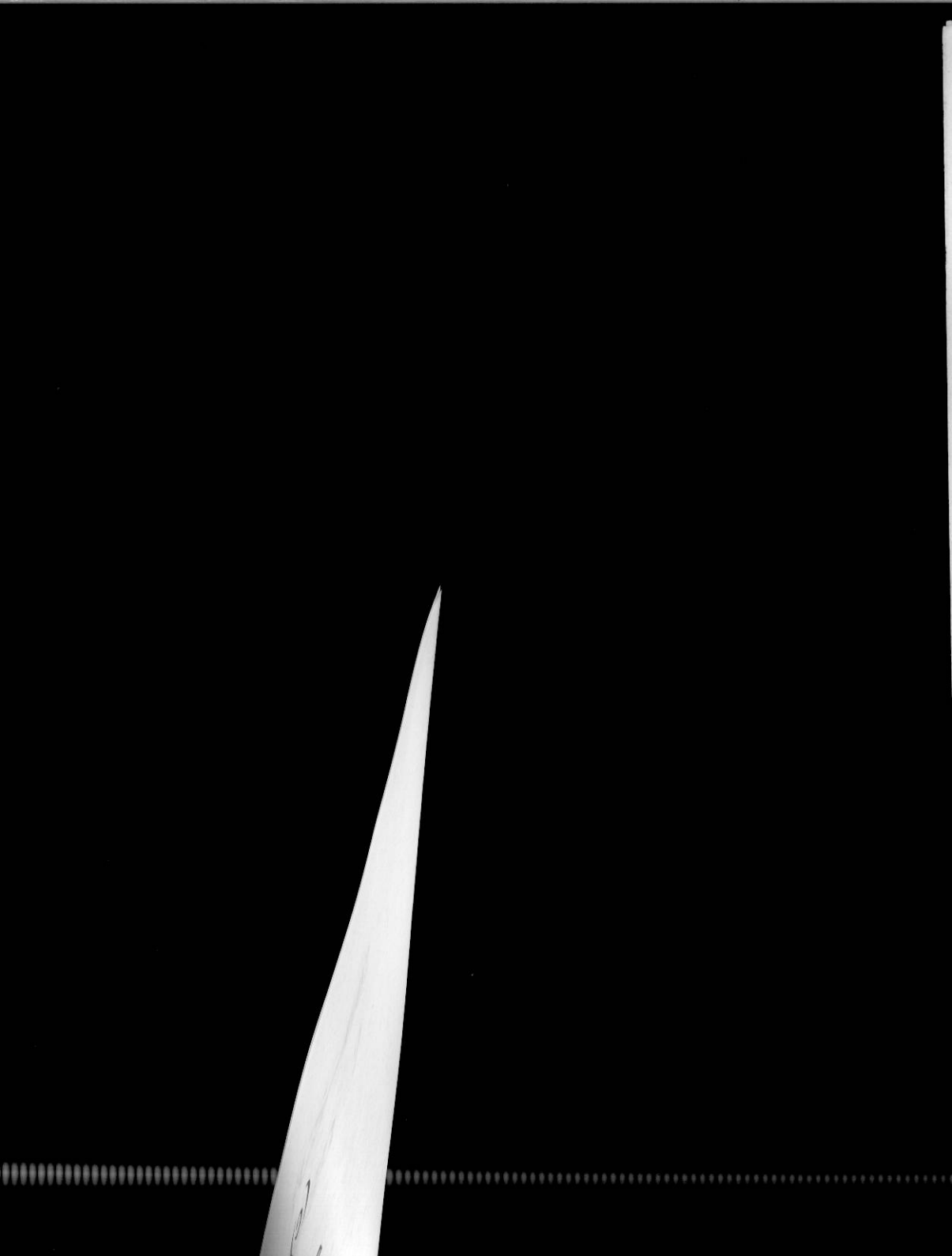
(b) List and describe the characteristics of software.

(c) Explain Incremental Model with the help of suitable example.

4. Attempt any *two* parts of choice from (a), (b) and (c). (5×2=10 Marks)

(a) Give the techniques used for Requirement Elicitation and discuss pros and cons of each.

(b) Explain the process of project management using Scrum in Agile software development.



(c) Write short notes on Data Dictionary and Decision Table.

5. Attempt any *two* parts of choice from (a), (b) and (c). (5×2=10 Marks)

(a) Explain the process of Requirement Engineering in detail.

(b) Explain the different phases in RUP Model and list the work flows associated.

(c) What is DFD ? Design the various levels of DFD for Hostel Management System.

Roll No.

TCS-704

**B. Tech. (CS/IT)
(SEVENTH SEMESTER)**

MID SEMESTER EXAMINATION, 2019

ADVANCED COMPUTER ARCHITECTURE

Time : 1 : 30 Hours

Maximum Marks : 50

Note :(i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Differentiate computer organization and computer architecture in brief.

(b) List different performance metrics and evaluation factors of CPU.

(c) State and prove Ahmdal's law.

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain Flynn's classification with neat diagram.
 - (b) Write short notes on any *two* of the following :
 - (i) Moore's law
 - (ii) Iron's law of performance
 - (iii) Lhadma's law
 - (c) What are the different classes of parallelism and parallel architecture ?
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) What do you understand by Memory Hierarchy ?
 - (b) Consider a direct mapped cache with block size 4 kB. The size of main memory is of 16 GB and there are 10 bits in the tag. Compute size of cache memory, tag directory size.
 - (c) State and explain the types of miss in the cache.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Define spatial and temporal locality of reference.

(3)

- (b) Consider a direct mapped cache with 8 cache blocks (0 – 7). If the memory block requests are in the order :

3, 5, 2, 8, 0, 6, 3, 9, 16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82, 17, 24.

What are the memory blocks that will be present in cache at the end of the sequence ? Also calculate Hit and Miss ratio.

- (c) Differentiate serial processing and parallel processing with suitable example.

5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) What is the need of virtual memory ? Explain in detail.

- (b) Suppose that we are considering an enhancement that runs 10 times faster than the original machine but is usable only 40% of the time. What is the overall speedup gain by incorporating the enhancement ?

- (c) What is cache coherence ? How can we get ride the problem of cache coherence ?

9.30 to 11 Am

Roll No.

08/12/19

TIT-704

B. TECH. (CS/IT)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019
CRYPTOGRAPHY AND NETWORK
SECURITY

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any two subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are ten.

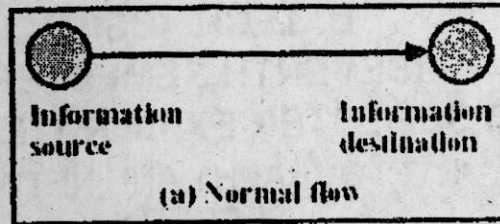
1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Explain the following concepts using appropriate examples to justify your answer :

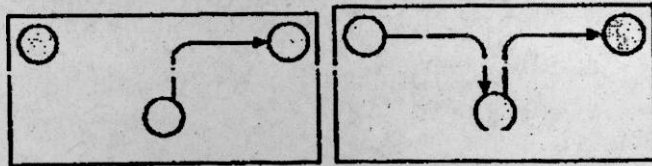
(i) Cryptography–Applications and algorithms

(ii) Network Security–Applications and technologies

- (b) A (source) is sending message to B (destination) through Internet. Normal flow is shown below :



However, due to potential threat, following attacks have observed to be experienced on the message during transmission :



- In each of the case (i) Identify the attack along with proper explain about the implications and (ii) Define the service that need to be incorporated to overcome this attack.
- (c) Briefly explain the concept of Caesar Cipher. You intercept the following ciphertxt :
- GTRG UQFSYJI FY YMJ MNLMBFD
SZRGJW TSJ

You also analyze that this ciphertext is generated using a Caesar Cipher and the key used is between 3 to 6. Can you identify exact key and find out meaningful plaintext ?

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) What is meant by Security Service ?
Explain various security services listed in X.800.

(b) Briefly discuss the concept of Vigenere cipher along with the key space available and possible approaches to attack the ciphertext. Further consider one possible combination of the Vigenere Tableau given below :

	a	b	c	d	e	f
A	A	B	C	D	E	F
B	B	C	D	E	F	A
C	C	D	E	F	A	B
D	D	E	F	A	B	C
E	E	F	A	B	C	D
F	F	A	B	C	D	E

Using the keyword "FACE" encrypt the plaintext "BAD ACES".

(c) What is the difference between an unconditionally secure cipher and a computationally secure cipher ? Explain the algorithm that is considered unconditionally secure in brief and emphasize its limitation for practical usage.

3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) How is Steganography different from Cryptography ? Provide explanation to concept of hiding data into Colour and Gray scale images.

(b) Given a colour (RGB) image having resolution of 1024 pixels by 720 pixels :

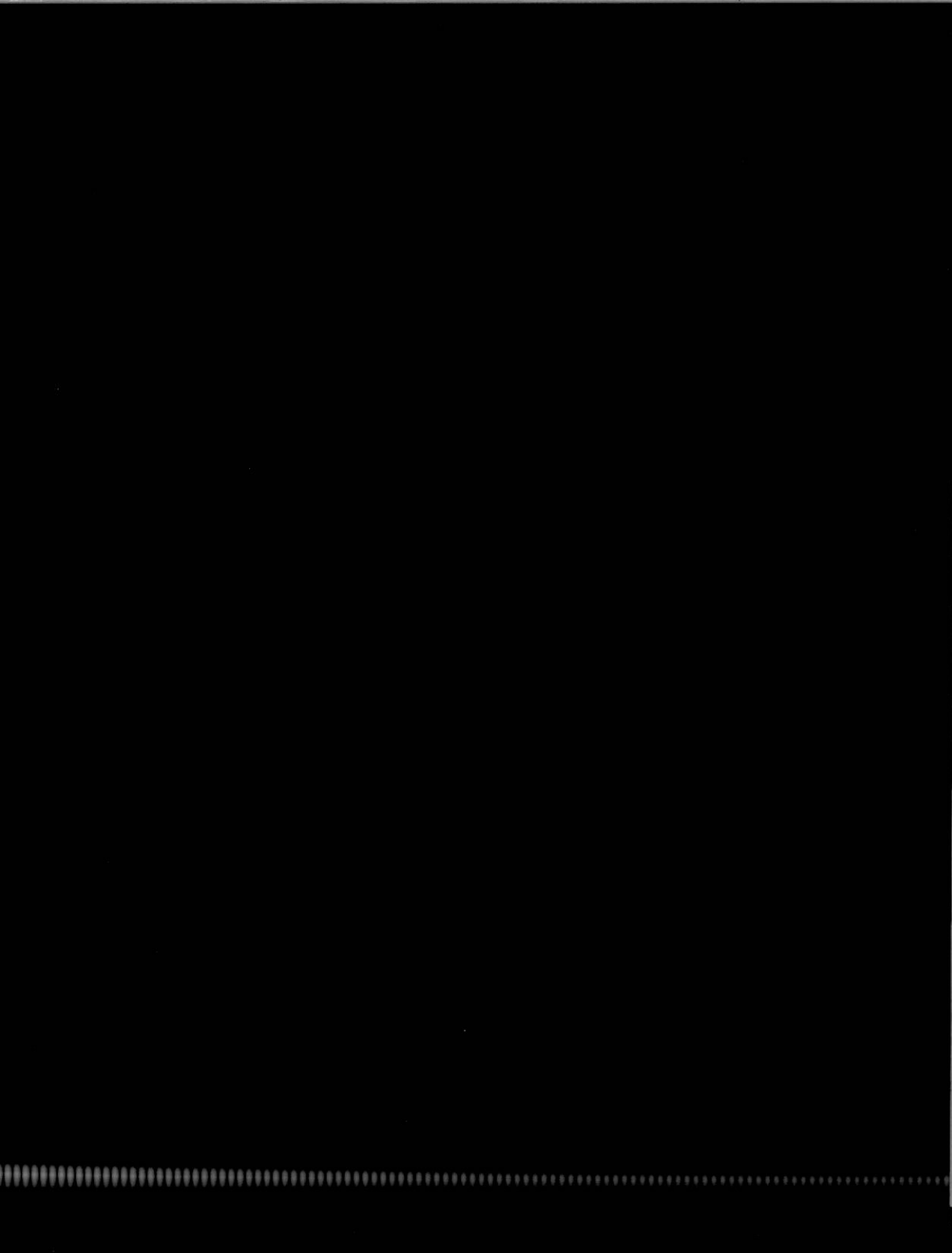
(i) How many bytes does this image constitute ?

(ii) One least significant bit of every byte is used for hiding the data. Maximum how many bytes of data can be sent using this image ? Justify your answer with detailed calculation.

- (c) Explain the concept of NULL cipher. Here is the intercepted message. It makes use of NULL cipher. Can you identify the hidden message ?

Greetings to all at Oxford. Many thanks for your letter and for the Summer examination package. All Entry Forms and Fees Forms should be ready for final despatch to the syndicate by Friday 20th or at the very latest, I'm told, by the 21st. Admin has improved here, though there's room for improvement still; just give us all two or three more years and we'll really show you! Please don't let these wretched 16t proposals destroy your basic O and A pattern. Certainly this sort of change, if implemented immediately, would bring chaos.

4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) With the help of a neat figure, explain the details of round function 'fk' of S-DES.
- (b) What is the reason for introduction of confusion and diffusion in the Block cipher ? Explain confusion and diffusion in detail and state which components/blocks are used in block cipher to achieve these.



- (c) Discuss the Linear Congruential Algorithm for generating the random numbers and mention criteria that are used for evaluating random number generator algorithm.

5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) How does S-DES perform encryption and decryption ? Explain with the help of neat block diagram.
- (b) What was the need of developing double DES ? With the help of a block diagram, explain the concept and strength of Double DES.
- (c) Consider transmission of message over the Internet. It is possible to secure message using either Link Level Encryption *or* End-to-end Encryption. Explain these concepts with the help of neat figure. Also identify merits and limitations of each of the approach.

25/9/12
9:30 to 11 Am

Roll No.

TCS-754

**B. TECH. (CSE)
(SEVENTH SEMESTER)**

**MID SEMESTER EXAMINATION, 2019
MOBILE APPLICATION DEVELOPMENT**

Time : 1 : 30 Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) What is android ecosystem ?

(b) What languages are supported by Android for application development ?

(c) Explain file structure of Andoid studio used for mobile application development.

(2)

TCS-754

2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
 - (a) Compare the Native mobile application development with Hybrid application development with suitable examples.
 - (b) Describe the architecture of android operating system with suitable diagram.
 - (c) Differentiate between emulator and simulator.
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
 - (a) What is the difference between JVM and DVM ?
 - (b) What are Shared Preferences ? Explain, how we can access the shared preferences.
 - (c) What composes an Android application project developed using Cordova framework ?
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
 - (a) Explain the various storage provided by the android.
 - (b) Discuss the deployment issues of mobile applications in marketplace.

(3)

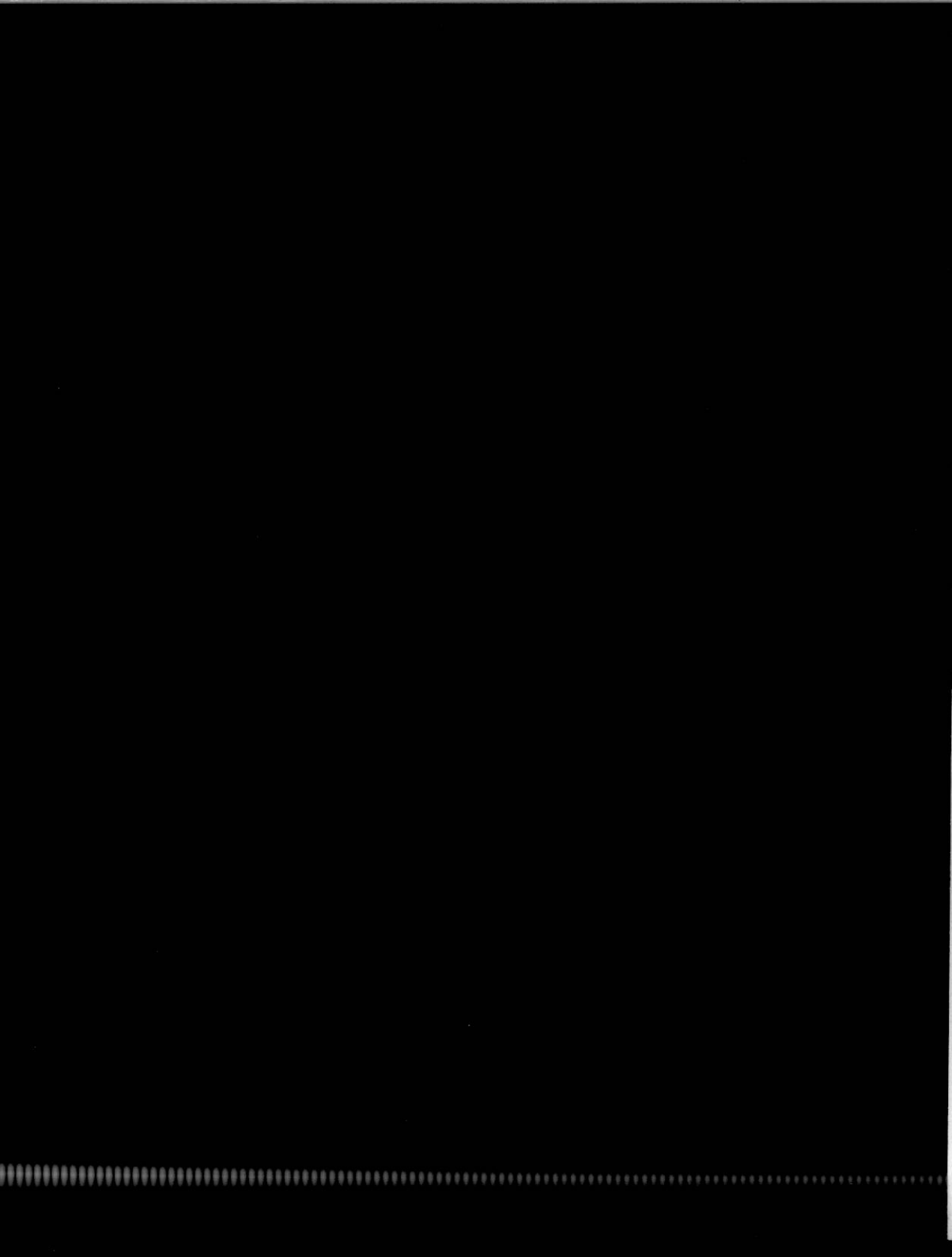
(c) Difference between Service and Internet Service.

5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) What is the importance of settings permissions in app development ?

(b) What's the difference between onCreate() and onStart() ?

(c) Write down the steps for creating Cordova application.



Roll No.

TCS-703

**B. TECH. (CS/IT)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019
COMPUTER NETWORKS—II**

Time : 1 : 30 Hours

Maximum Marks : 50

Note :(i) All questions are compulsory.

(ii) Answer any *two* subquestions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Differentiate between ALOHA and slotted ALOHA.

(b) Use module 2 arithmetic to compute the CRC code for message 101101110011 with the generator 10011.

- (c) One node transmits 100 byte frames at 10 frames per second, another transmits 1000 byte frames at 2 frames per second, calculate the utilization of the Ethernet LAN. Assume 10 Mbps. Show all your work.
2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain the Ethernet Frame Structure.
 - (b) Differentiate between Switches and Routers.
 - (c) The Standard Ethernet (CSMA/CD) protocol uses exponential backoff. Describe the Ethernet exponential backoff algorithm (your explanation must include why it is called exponential).
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Show (give an example of) that two-dimensional parity checks can correct and detect a single bit error. Show (give an example of) a double-bit error that can be detected but not corrected.

- (b) Consider the simple networking scheme given below :

HostA---SwitchA----Router-----SwitchB
-----HostB

||

||

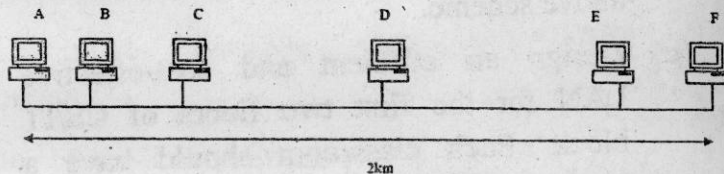
HostCHostD

Explain the working of ARP when HostA wants to send a message to HostB in the above scheme.

- (c) Design an efficient and cost-effective LAN for the first two floors of CS/IT block. Each classroom should have a connection, LT-1, 2, 3, 4, 5 and conference room and Seminar Halls will also have projectors which will also be directly connected to the LAN. Decide suitable places for a departmental web server and a mail server. Faculty rooms will have single computers with printers. Decide a suitable place for a shareable printer on the LAN which can be used by the students. Location of switches etc. (whenever required) should be clearly marked along with its transmission rate and number of ports.

4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) What are the four desirable characteristics of a broadcast channel ? Which of these characteristics does slotted ALOHA have ? Which of these characteristics does token passing have ?
- (b) Consider the following linear network that is 2 km long :



Assume that propagation speed of electromagnetic waves sent over the medium is 2×10^8 m/s and the transmission rate of the network is 10 Mbps. What is the minimum frame size (in bits) necessary to ensure that CSMA/CD will work properly for this network ? Explain your reasoning.

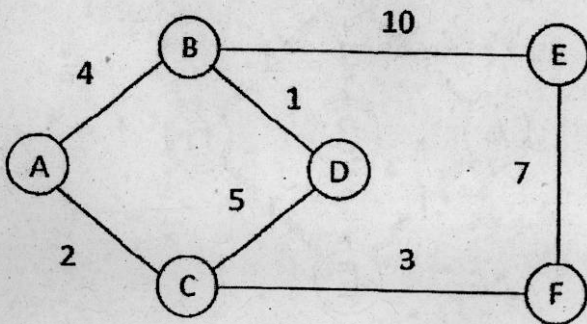
- (c) In CSMA/CD, after the fifth collision, what is the probability that a node chooses $K = 5$? The result $K = 5$ corresponds to a delay of how many seconds on a 100 Mbps Ethernet ?

(5)

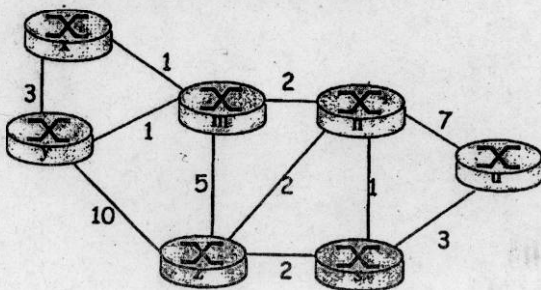
TCS-703

5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) Consider the network given below. The link between C and F fails after some time. Compute the routing table at D after the failure of the link using LS routing algorithm :



- (b) Consider the network given below. Run the LS algorithm to compute the shortest path from y to all network nodes. Show your work in tabular format :



- (c) Consider the network shown below and assume that each node initially knows the costs to each of its neighbours. Use the distance vector algorithm to complete the entire distance table below as it would look like at every node after the algorithm has converged :

